Autonomous AI-Driven Computational Search for Three Mutually Orthogonal Latin Squares of Order 10: The Autoresearch Agentic Framework Approach

Research Report

Autoresearch Project (Claude AI Agents)
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Abstract

We present a systematic computational investigation of the sixty-year-old open problem of whether three mutually orthogonal Latin squares (3-MOLS) of order 10 exist, i.e., whether $N(10) \geq 3$. Our work is distinguished by a novel methodology: an *autonomous agentic research framework* in which Claude AI agents independently design algorithms, write code, launch parallel compute jobs, monitor results, and update research strategy across 30 multi-week sessions without human intervention.

The agents developed and iteratively refined a suite of algorithms: CP-SAT hint-guided cascade optimization (OR-Tools), large neighborhood search (LNS), SA-CT climbing for pair generation, a near-miss pool with warm-start hints, BFS depth-5 intercalate neighborhood analysis, and feasibility attacks with hard energy constraints. These tools were applied to a catalog of 323+ verified MOLS-10 pairs generated by five construction methods.

Our main findings: (1) The minimum total clash energy $\mathcal{E} = \mathrm{cl}_{13} + \mathrm{cl}_{23}$ cascaded $37 \to 34 \to 33 \to 32 \to 31 \to 30 \to 29 \to 26 \to \mathbf{23}$ across 32+ sessions; the current all-time record is $\mathcal{E} = \mathbf{23}$ ($\mathrm{cl}_{13} = 9$, $\mathrm{cl}_{23} = 14$) for pair `tv_254`, found at Session 32 via 8-replica *Aggressive Parallel Tempering* with temperature range $T \in [0.3, 5.0]$ (geometrically spaced, 300k steps/round), achieving 50–95% swap rates throughout (no bottleneck). Previous record was $\mathcal{E} = 26$ for pair `iso_tv_21_0` (fingerprint `ee29e34e`), found at Session 26b via 4-worker CP-SAT hint cascade. (2) Structural analysis proves $\mathcal{E} = 23$ is a *strict 4-deep intercalate (IC) local minimum*: exhaustive BFS to depth 4 finds 28,553 unique IC-reachable states, all with $\mathcal{E} \geq 23$ (depth-1 min = 26, depth-2 min = 26, depth-3 min = 25, depth-4 min = 25). IC-only SA cannot escape this basin; escape requires non-IC moves (row/column permutations or symbol relabelings of L_3). For the prior record, structural analysis proves $\mathcal{E} = 26$ (`iso_tv_21_0`) is a strict local minimum under all standard 1-move types; a 7200-worker-second feasibility attack with constraint $\mathcal{E} \leq 25$ finds 0 SAT and 0 INFEASIBLE (all timeout). (3) **Three structurally independent MOLS-10 pairs all reach** $\mathcal{E} = 28$: the secondary cascade of `iso_tv_21_0` ($\mathrm{cl}_{13} = 16$, $\mathrm{cl}_{23} = 12$), `mdecomp_279` (fingerprint `e54e791c`, $\mathrm{cl}_{13} = 8$, $\mathrm{cl}_{23} = 20$), and `mdecomp_113` (fingerprint `2469c6dc`, $\mathrm{cl}_{13} = 17$, $\mathrm{cl}_{23} = 11$), suggesting a structural platform in the energy landscape at $\mathcal{E} = 28$. (4) Every one of the 344 originally tested MOLS-10 pairs has $\mathrm{CT}(L_1, L_2) \leq 2 < 10$, with Glucose3 UNSAT certificates for all 323 pairs (≈ 42 ms each), initially suggesting a *CT barrier conjecture* ($\mathrm{CT} \leq 2$ for all pairs, which would imply $N(10) = 2$). (5) A structural reformulation — a 3-MOLS(10) exists iff a single square admits two mutually orthogonal transversal decompositions — redirected the search to transversal-rich squares. Exhaustive CP-SAT censuses show the catalog squares admit at most 4 decompositions each, none mutually orthogonal, and the $\mathcal{E} = 23$ record square has a *unique* mate with $\mathrm{CT} = 0$. Pivoting to the 2^{25} *turn-square* family (which attains the order-10 maximum of 5504 transversals) and streaming orthogonal mates via CP-SAT solution enumeration **refutes the strong CT barrier conjecture**: a turn-square mate pair

with CT = 6 was found and independently verified. Mate-space large-neighborhood search saturates at CT = 6 over 45.8M steps; exact max-CT optimization over the full mate space is formulated (via an intersection-reduction lemma cutting the model to 26.5% of pairs) and running.

Collectively, the energy-cascade and census results constitute strong computational evidence for $N(10) = 2$, while the turn-square CT breakthrough shows the pair landscape is richer than the early data suggested; the question remains formally open.

Contents

1 The N(10) MOLS Problem: Introduction and Challenges

1.1 Latin Squares and Mutual Orthogonality

Definition 1.1 (Latin square)**.** *A* Latin square *of order n is an $n \times n$ array L with entries from the symbol set $[n] = \{0, 1, \ldots, n-1\}$ such that each symbol appears exactly once in every row and exactly once in every column.*

Definition 1.2 (Orthogonal Latin squares)**.** *Two Latin squares L_1 and L_2 of order n are* orthogonal*, written $L_1 \perp L_2$, if the n^2 ordered pairs $(L_1(i,j), L_2(i,j))$ are all distinct as (i,j) ranges over the n^2 cells. Equivalently, every ordered pair $(a,b) \in [n]^2$ appears in exactly one cell of the superposition.*

Definition 1.3 (MOLS and $N(n)$)**.** *A set $\{L_1, L_2, \ldots, L_k\}$ of k Latin squares of order n is called a set of* mutually orthogonal Latin squares *(*MOLS*) if $L_i \perp L_j$ for all $i \neq j$. The maximum size k of any such set is denoted $N(n)$.*

When $n = q$ is a prime power, the classical Galois field construction yields $N(q) = q - 1$. For example, $N(7) \geq 6$, $N(8) \geq 7$, $N(9) \geq 8$. In general, the MacNeish bound [4] gives $N(n) \leq n - 1$ for all n.

The question of $N(10)$ is fundamentally harder because $10 = 2 \times 5$ is not a prime power, so no direct algebraic construction applies.

1.2 Historical Background

Euler's conjecture (1782). Euler [1] conjectured that $N(n) = 1$ (i.e., no two orthogonal Latin squares exist) for all $n \equiv 2 \pmod 4$, equivalently for $n = 2, 6, 10, 14, 18, \ldots$ This conjecture was motivated by the "36 Officers Problem": can 36 officers of 6 different ranks from 6 different regiments be arranged in a 6×6 square so that no rank or regiment is repeated in any row or column?

Tarry's verification for $n = 6$ (1901). Tarry [2] proved by exhaustive case analysis that $N(6) = 1$, confirming Euler's conjecture for $n = 6$. This result — the impossibility of arranging the 36 officers — remains one of the earliest large-scale computational combinatorics results.

Parker's disproof for $n = 10$ (1960). Parker [3] dramatically disproved Euler's conjecture by constructing two mutually orthogonal Latin squares of order 10, establishing $N(10) \geq 2$. In the same year, Bose, Shrikhande, and Parker [5] proved that $N(n) \geq 2$ for all $n \neq 2, 6$, completely refuting Euler's conjecture.

Lam's nonexistence result (1989). The upper bound $N(10) \leq 8$ comes from a landmark computational result: Lam, Thiel, and Swiercz [6] proved that no projective plane of order 10 exists. Since a projective plane of order n is equivalent to $n - 1$ MOLS of order n, nonexistence of a projective plane of order 10 yields $N(10) \leq 8$. The computation consumed tens of thousands of CPU hours.

Current state. The best known bounds are:

$$2 \leq N(10) \leq 8.$$

The problem of determining $N(10)$ precisely has been open for over 60 years. Most combinatorialists conjecture $N(10) = 2$ [7, 8], but no proof of this conjecture exists. In particular, the existence of a third MOLS of order 10—whether $N(10) \geq 3$—is completely unknown.

1.3 Mathematical Framework

1.3.1 The Clash Count and Energy Function

Definition 1.4 (Clash count)**.** *For Latin squares A and B of order n, the* clash count *is*

$$\mathrm{cl}(A,B) = n^2 - |\{(A(i,j), B(i,j)) : i,j \in [n]\}|,$$

i.e., the number of ordered pairs $(a,b) \in [n]^2$ that do not *appear in the superposition of A and B. We have $\mathrm{cl}(A,B) = 0$ if and only if $A \perp B$.*

For a triple (L_1, L_2, L_3) with $L_1 \perp L_2$ (so $\mathrm{cl}_{12} = 0$), we define the *energy*:

$$\mathcal{E}(L_1, L_2, L_3) = \mathrm{cl}(L_1, L_3) + \mathrm{cl}(L_2, L_3) = \mathrm{cl}_{13} + \mathrm{cl}_{23}.$$

The **goal** is to find (L_1, L_2, L_3) with $\mathcal{E} = 0$, which would prove $N(10) \geq 3$.

1.3.2 Common Transversals

Definition 1.5 (Common transversal)**.** *Let L_1, L_2 be orthogonal Latin squares of order n. A* common transversal *(CT) of the pair (L_1, L_2) is a set $T = \{(r_0, c_0), \ldots, (r_{n-1}, c_{n-1})\}$ of n cells satisfying:*

(i) each row index $0, \ldots, n-1$ appears exactly once;
(ii) each column index $0, \ldots, n-1$ appears exactly once;
(iii) the multiset $\{L_1(r_i, c_i)\}_{i=0}^{n-1}$ consists of all n distinct symbols;
(iv) the multiset $\{L_2(r_i, c_i)\}_{i=0}^{n-1}$ consists of all n distinct symbols.

We write $\mathrm{CT}(L_1, L_2)$ for the maximum number of pairwise disjoint common transversals of (L_1, L_2).

The fundamental connection between common transversals and MOLS is:

Theorem 1.6 (CT Necessity)**.** *A Latin square L_3 of order n satisfies $L_3 \perp L_1$ and $L_3 \perp L_2$ if and only if the n^2 cells can be partitioned into n pairwise disjoint common transversals $T_0, \ldots, T_{n-1}$ of (L_1, L_2), where $T_k = \{(i,j) : L_3(i,j) = k\}$.*

Proof sketch. Each symbol k of L_3 occupies exactly one cell per row and one per column (Latin square property). If $L_3 \perp L_1$, those n cells contain each L_1-symbol exactly once; if $L_3 \perp L_2$, they contain each L_2-symbol exactly once. Hence the n cells for symbol k form a common transversal T_k. The n transversals $T_0, \ldots, T_{n-1}$ partition all n^2 cells. The converse is direct. □

Corollary 1.7 (CT Barrier)**.** *If $\mathrm{CT}(L_1, L_2) < n$, then no Latin square L_3 satisfying $L_3 \perp L_1$ and $L_3 \perp L_2$ exists. In particular, for $n = 10$: if $\mathrm{CT}(L_1, L_2) \leq 9$, the pair (L_1, L_2) cannot be extended to three MOLS.*

1.3.3 SAT Encoding

The existence of $L_3 \perp L_1$ and $L_3 \perp L_2$ for fixed (L_1, L_2) is encoded as a SAT instance with 1000 Boolean variables $x_{i,j,k} = 1 \iff L_3(i,j) = k$, plus five families of exactly-one constraints enforcing the Latin square property of L_3 and orthogonality with both L_1 and L_2. An UNSAT certificate from a CDCL solver (Glucose3 [10]) definitively proves no L_3 exists for that pair.

1.4 Key Challenges

The $N(10)$ problem poses a combination of mathematical and computational challenges that have stymied researchers for over six decades:

1. **Exponential search space.** The number of distinct Latin squares of order 10 is approximately 10^{38} [16]. Even restricting to valid L_3 given fixed (L_1, L_2), the search space is vast.

2. **Local minima barriers.** Simulated annealing over the energy $\mathcal{E}$ reaches a floor of $\mathcal{E} = 37$ that cannot be escaped by standard local moves (row/col swaps, intercalates). This floor is a strict local minimum under all neighborhood moves tested prior to the introduction of CP-SAT optimization.

3. **The CT barrier.** All 344 MOLS-10 pairs tested in this project have $\mathrm{CT}(L_1, L_2) \leq 2$, far below the required threshold of $n = 10$. By Corollary 1.7, none of these pairs can yield 3-MOLS. Whether *any* MOLS-10 pair can achieve $\mathrm{CT} \geq 10$ is unknown.

4. **No projective plane.** The nonexistence of a projective plane of order 10 rules out the construction of all $n - 1 = 9$ MOLS simultaneously, but gives no direct obstacle to finding 3 MOLS.

5. **Absence of algebraic structure.** Since 10 is not a prime power, there is no Galois field GF(10), and the standard algebraic MOLS construction does not apply. All known approaches rely on combinatorial search.

6. **Scale of prior computation.** Lam et al. [6] required tens of thousands of CPU hours to settle the related question of the projective plane. A direct exhaustive search for $N(10) \geq 3$ appears computationally infeasible with current resources without major algorithmic advances.

2 The Autoresearch Agentic Framework

2.1 Framework Overview

The *Autoresearch Agentic Framework* is an autonomous AI research system in which Claude-powered agents conduct end-to-end computational research without human intervention. Over 30 sessions spanning multiple weeks, agents executed the following research cycle repeatedly:

1. **State assessment.** Read prior session logs, check the near-miss pool, and summarize current best energies and which pairs have been explored.

2. **Strategy decision.** Based on the current state, decide which algorithms to run, which pairs to target, and what compute budget to allocate.

3. **Implementation.** Write or modify Python/shell code for new algorithms, fix bugs identified in prior sessions, and configure parameters.

4. **Execution.** Launch parallel worker processes, monitor output, kill stalled workers, and save breakthroughs to the near-miss pool.

5. **Analysis.** Interpret results, update the research direction, and document findings for the next session.

This cycle ran in a **15-minute autonomous loop**: within each session, agents continuously checked progress and acted without waiting for human feedback.

2.2 Key Agent Capabilities

The framework enabled capabilities that would be impractical for human researchers working alone:

- **Parallel process management.** Agents spawned dozens of parallel search processes (each using 2–4 CPU workers), monitored their output in real time, killed processes that stalled at local minima, and restarted them with new configurations.
- **Adaptive exploration.** Agents tracked which of the 323+ MOLS-10 pairs had been explored and at what energy, then allocated compute preferentially to the most promising pairs.
- **Breakthrough detection.** When a new energy record was found, the agent immediately updated the near-miss pool, used the new L_3 as a hint for subsequent CP-SAT runs (the hint cascade), and shifted other workers to explore from the new record.
- **Algorithm invention.** Key algorithmic innovations — the hint cascade strategy, the BFS depth-5 local minimum proof, the 4-row LNS analysis, and the feasibility attack — were all conceived and implemented by agents during the course of the research.
- **Cross-session continuity.** Agents persisted state (near-miss pool, pair catalog, best-energy files) across sessions, enabling long-running cascades to span multiple autonomous sessions.

2.3 Algorithm Portfolio

Agents developed and deployed the following suite of algorithms:

2.3.1 CP-SAT Hint-Guided Optimization

The core optimization engine uses Google OR-Tools CP-SAT [9], a constraint-programming solver with integrated SAT and linear programming. For fixed (L_1, L_2) from the near-miss pool, the model minimizes $\mathcal{E} = \mathrm{cl}_{13} + \mathrm{cl}_{23}$ subject to L_3 being a valid Latin square orthogonal to neither but close to both.

Key design choices:

- Decision variables: 100 integer variables $L_3[i, j] \in [10]$.
- Orthogonality coverage indicators: 200 Boolean variables, one per ordered pair (a, b) for each of cl_{13} and cl_{23}.
- Symmetry breaking: fixing L_3's first row to the identity reduces search by factor 10! without loss of generality.
- Parallel workers: 2–4 CP-SAT workers running portfolios of different heuristics.

2.3.2 Hint Cascade Strategy

The agents' most impactful innovation is the *hint cascade*: a chain of CP-SAT runs where each run receives the best L_3 from the previous run as a warm-start hint.

The cascade allows CP-SAT to leverage its non-local branching ability at each step while being guided toward regions already known to be good.

Algorithm 1 Hint Cascade for Pair (L_1, L_2)

Require: MOLS pair (L_1, L_2) with $\mathrm{cl}_{12} = 0$, timeout T, max rounds K
Ensure: Best L_3 and energy $\mathcal{E}^*$ found
$L_3^{(0)} \leftarrow$ random Latin square; $\mathcal{E}^{(0)} \leftarrow \mathcal{E}(L_1, L_2, L_3^{(0)})$
$E^* \leftarrow \mathcal{E}^{(0)}$; $L_3^* \leftarrow L_3^{(0)}$
for $k = 1, 2, \ldots, K$ **do**
  Run CP-SAT with hint $L_3^{(k-1)}$, minimize $\mathcal{E}$, timeout T
  Obtain $(L_3^{(k)}, \mathcal{E}^{(k)})$
  if $\mathcal{E}^{(k)} < E^*$ **then**
    $E^* \leftarrow \mathcal{E}^{(k)}$; $L_3^* \leftarrow L_3^{(k)}$
  end if
  if $\mathcal{E}^{(k)} = 0$ **then return** L_3^*, E^* ▷ 3-MOLS found!
  end if
  if $\mathcal{E}^{(k)} \geq \mathcal{E}^{(k-1)}$ **then break** ▷ No improvement
  end if
end for
return L_3^*, E^*

2.3.3 MOLS Pair Generation Methods

Five methods were used to generate the catalog of 323+ verified MOLS-10 pairs:

1. **Transversal Decomposition (TV).** Fix Parker's TV-21 square and enumerate its transversals; partition cells into 10 disjoint transversals via Dancing Links [12] to obtain orthogonal mates.

2. **Isotopy Variants (ISO).** Apply row permutations, column permutations, and symbol relabelings to known MOLS pairs to generate new pairs in different isotopy classes.

3. **SA-CT Climbing.** Run simulated annealing [11] with the objective of maximizing $\mathrm{CT}(L_1, L_2)$ while maintaining $\mathrm{cl}_{12} = 0$.

4. **Modular Decomposition (mdecomp).** Construct MOLS pairs from modular arithmetic and transversal decomposition of random Latin squares; generates pairs in novel L_1-families not isotopic to Parker's original.

5. **Intercalate Chain (IC).** Apply bounded sequences of intercalate flips to known pairs to generate nearby pairs.

2.3.4 Additional Search Methods

- **SA warm-starts.** Simulated annealing initialized from a near-miss pool entry, allowing the energy to be pushed below the SA floor of $\mathcal{E} = 37$.

- **Large Neighborhood Search (LNS).** Fix a subset of k rows (or columns) of L_3 as free variables; re-optimize using CP-SAT. Agents tested all $\binom{10}{4} = 210$ four-row subsets to prove 4-row local optimality.

- **BFS Intercalate Analysis.** Breadth-first search over all intercalate flips from a given L_3, up to depth d, to certify that no sequence of d intercalate moves can reduce $\mathcal{E}$.

- **Feasibility Attack.** Run CP-SAT with the hard constraint $\mathcal{E} \leq E_{\mathrm{target}}$ (rather than minimizing $\mathcal{E}$). Returns SAT (found better solution), UNSAT (proved lower bound $\mathcal{E} > E_{\mathrm{target}}$), or TIMEOUT.

- **Glucose3 SAT Certificates.** For each MOLS-10 pair, encode the existence of L_3 as a CNF instance and run Glucose3 [10]. An UNSAT result certifies $\mathrm{CT}(L_1, L_2) < 10$.

2.4 The Near-Miss Pool

All triples (L_1, L_2, L_3) with energy $\mathcal{E} \leq 37$ are stored in a persistent near-miss pool. The pool serves two roles:

1. **Hint source.** The best pool entries provide L_3 hints for CP-SAT cascades.
2. **Progress tracker.** The minimum pool energy is the global benchmark; agents report it at the start of each session.

File-locking (`fcntl.LOCK_EX`) ensures race-condition-free writes when multiple parallel workers update the pool simultaneously.

3 Stage-by-Stage Agent Results

3.1 Overview: The Energy Cascade

Table 1 summarizes the energy cascade achieved across all 30 sessions.

Table 1: Energy cascade across sessions: best $\mathcal{E}$ per milestone.

Sessions	Key Development	Best $\mathcal{E}$	Pair
1–8	SA floor established; 323+ pairs generated	37	various
9	CP-SAT breaks SA floor (2 independent pairs)	**34**	various
10–11	SA warm-start cascade: $34 \to 33 \to 32 \to 31$	**31**	various
12–14	CP-SAT cascade: $31 \to 30 \to 29$	**29**	4f04a773 class
14b	2-intercalate neighborhood: $30 \to 29$ instantly	**29**	4f04a773 class
15–16	4 isotopy classes reach E=29; BFS depth-5 proofs	**29**	4 classes
17–25	Full 323-pair scan; iso_tv_21_0 scan record E=32	**29**	ee29e34e
25b	iso_tv_21_0 v3 trial 1: cold **E=30** (new record)	**29**	ee29e34e
26	iso_tv_21_0 v3 trial 2: hint cascade **E=29**	**29**	ee29e34e
26b	iso_tv_21_0 v3 trial 4: **E=26 ALL-TIME RECORD**	**26**	ee29e34e
27–28	E=26 structural analysis; feasibility attack	**26**	ee29e34e
29–30	mdecomp_279 and mdecomp_113 reach **E=28**	**26**	multiple
31	Fast numba SA; iso2_tv21_0_5 & tv_254 reach $\mathcal{E} = 26$ via cascade	**26**	multiple
32	Aggressive PT ($T \in [0.3, 5.0]$): tv_254 **E=23 ALL-TIME RECORD**	**23**	tv_254
32+	BFS confirms E=23 is strict 4-deep IC local min; escape SA ongoing	**23**	tv_254

3.2 Stage 1 (Sessions 1–8): Infrastructure and Pair Generation

3.2.1 Pair Catalog Construction

Agents constructed a catalog of 323 verified MOLS-10 pairs using the five generation methods described in Section 2. Table 2 shows the breakdown by method.

3.2.2 The CT Barrier: First Evidence

For all 323 pairs, Glucose3 UNSAT certificates confirmed $\mathrm{CT}(L_1, L_2) \leq 2$. By Corollary 1.7, none of these pairs can be extended to 3-MOLS.

Table 2: MOLS-10 pair catalog by construction method.

Method	Pairs	CT=1	CT=2	Notes
TV variants (Parker 1960)	56	46	10	Based on TV-21 square
Isotopy transforms (ISO)	22	8	14	Permuted TV-21 variants
Intercalate chains	6	3	3	IC depth 2–12
mdecomp	136	2	134	Novel L_1 families
SA-CT climbing	103	0	103	Random-start CT optimization
Total	**323**	**58**	**265**	All with $\mathrm{cl}_{12} = 0$

Table 3: CT distribution across 323 MOLS-10 pairs.

CT value	Pairs	Fraction
1	59	18.3%
2	264	81.7%
≥ 3	**0**	**0%**

3.2.3 Simulated Annealing Floor: $\mathcal{E} = 37$

A 7-replica parallel-tempering SA [11] searched for L_3 over all pairs. After hundreds of 97-second trials across 8 sessions, the SA floor was firmly established at $\mathcal{E} = 37$. This is a *strict local minimum* under all SA move types (row swap, column swap, symbol relabel, intercalate, k-scramble, big-scramble): no move from any $\mathcal{E} = 37$ state leads to $\mathcal{E} < 37$.

The $\mathcal{E} = 37$ floor held across all pools of starting states, all pairs, and all random seeds. This indicated that breaking the floor required a non-local optimization method.

3.3 Stage 2 (Sessions 9–16): First Cascades, $\mathcal{E} = 37 \to 29$

3.3.1 Session 9: CP-SAT Breaks the SA Floor

The introduction of OR-Tools CP-SAT in Session 9 immediately broke the SA floor:

Table 4: Session 9 CP-SAT results: breaking $\mathcal{E} = 37$.

Run	cl_{13}	cl_{23}	$\mathcal{E}$	Time	Workers	Notes
No symbreak, trial 2	18	18	36	180 s	2	First CP-SAT improvement
Symbreak + hint($\mathcal{E}$=39)	17	17	**34**	300 s	4	Two independent pairs
Second pair	18	16	**34**	120 s	2	Independently confirmed

Why CP-SAT succeeds where SA fails. SA moves change one or a few cells at a time; from $\mathcal{E} = 37$, every such move increases energy. CP-SAT's branch-and-bound with constraint propagation can make large, non-local assignments. With symmetry breaking reducing redundancy by a factor of $10! \approx 3.6 \times 10^6$, CP-SAT navigates directly to regions unreachable by local SA moves.

The fact that two independent (L_1, L_2) pairs both reached $\mathcal{E} = 34$ within the same session ruled out pair-specific luck.

3.3.2 Sessions 10–14: Cascade $\mathcal{E} = 34 \to 29$

Armed with $\mathcal{E} = 34$ pool entries, SA warm-starts immediately descended further. The agents then alternated SA warm-starts (fast descent) with CP-SAT hint runs (deeper optimization) in a cascade:

- Session 10: SA warm-start from $\mathcal{E} = 34$ pool $\to \mathcal{E} = 33$ ($\text{cl}_{13} = 14$, $\text{cl}_{23} = 19$). File-locking race condition identified and fixed.
- Session 10–11: CP-SAT cascade: $\mathcal{E} = 33 \to 32 \to 31$.
- Session 12–13: Further CP-SAT optimization: $\mathcal{E} = 31 \to 30$.
- Session 14: CP-SAT proof worker finds $\mathcal{E} = 30 \to 29$ in 311 seconds.
- Session 14b: A 2-intercalate neighborhood search from the $\mathcal{E} = 30$ state *immediately* found $\mathcal{E} = 29$ ($\text{cl}_{13} = 15$, $\text{cl}_{23} = 14$), proving the intercalate neighborhood of $\mathcal{E} = 30$ contains $\mathcal{E} = 29$ states.

3.3.3 Session 16: Four Isotopy Classes Proved at $\mathcal{E} = 29$

By Session 16, four distinct isotopy classes had reached $\mathcal{E} = 29$:

Table 5: The four isotopy classes at $\mathcal{E} = 29$ (Session 16).

Class fingerprint	cl_{13}	cl_{23}	IC count	BFS depth proved
4f04a773	15	14	28	5 (22M+ states)
2cc8e22f	14	15	23	5 (22M+ states)
74bb7b86	17	12	19	5 (22M+ states)
0968fc79	17	12	24	5 (22M+ states)

The BFS depth-5 analysis (over 22 million states each) proved that no sequence of up to 5 intercalate flips from any $\mathcal{E} = 29$ solution leads to $\mathcal{E} < 29$. These are *global intercalate-neighborhood minima at depth 5*.

Structural observation (classes 74bb7b86 and 0968fc79). These two classes share the same L_1 matrix (Hamming distance 0/100) but differ in L_2 (Hamming distance 50/100). Remarkably, all four cross-combinations (L_2, L_3) yield exactly $\mathcal{E} = 29$ with identical clash distributions, revealing a deep structural symmetry in the energy landscape.

3.4 Stage 3 (Sessions 17–26b): Full Scan and $\mathcal{E} = 26$ Record

3.4.1 Complete 323-Pair Scan

Sessions 17–25 performed a systematic cold-start scan of all 323 MOLS-10 pairs, allocating 120–180 seconds of CP-SAT optimization per pair. Key findings:

- 17 “standout” pairs reached $\mathcal{E} \leq 37$ in cold scan.
- iso_tv_21_0 (fingerprint `ee29e34e`, L_1 family #12) achieved the scan record $\mathcal{E} = 32$ ($\text{cl}_{13} = 13$, $\text{cl}_{23} = 19$), making it the top candidate for deep optimization.
- mdecomp_279 (fingerprint `e54e791c`) achieved co-scan-record $\mathcal{E} = 33$ ($\text{cl}_{13} = 12$, $\text{cl}_{23} = 21$), the first standout from a non-TV L_1 family.
- 28 distinct L_1 families were identified across all 323 pairs.
- The scan completed at Session 26b: all 323/323 pairs processed.

3.4.2 iso_tv_21_0: The Hint Cascade to $\mathcal{E} = 26$

Agents focused the hint cascade on iso_tv_21_0 (fingerprint `ee29e34e`), launching a dedicated “v3” push with 4 parallel CP-SAT workers:

The E=26 result ($\text{cl}_{13} = 8$, $\text{cl}_{23} = 18$, seed=1943590465): the cascade jumped from $\mathcal{E} = 29$ to $\mathcal{E} = 26$ in a single 4-worker CP-SAT trial. The trajectory $32 \to 30 \to 29 \to 26$ required only four trials total.

This is a drop of 3 energy units in a single step, demonstrating CP-SAT’s ability to make large non-local jumps in the energy landscape that are inaccessible to intercalate- based local search.

Table 6: iso_tv_21_0 v3 hint cascade achieving the E=26 all-time record.

Trial	Hint	cl_{13}	cl_{23}	$\mathcal{E}$	Notes
1 (cold)	none	15	15	30	New push co-record!
2	E=30 hint	14	15	29	seed=1333021984
3	E=29 hint	14	15	29	No improvement
4	E=29 hint	**8**	**18**	**26**	seed=1943590465, **ALL-TIME RECORD**

3.5 Stage 4 (Sessions 27–28): Structural Analysis of $\mathcal{E} = 26$

3.5.1 Proving $\mathcal{E} = 26$ is a Strict Local Minimum

Agents performed an exhaustive analysis of the 1-move neighborhood of the $\mathcal{E} = 26$ state:

Table 7: Exhaustive 1-move neighborhood of the E=26 state.

Move type	Neighborhood size	Min $\mathcal{E}$ observed	Result
Row swap	$\binom{10}{2} = 45$	43	No improvement
Column swap	$\binom{10}{2} = 45$	42	No improvement
Symbol relabel	$\binom{10}{2} = 45$	26	No improvement
Single intercalate	≤ 2500	28	No improvement

Theorem 3.1 ($\mathcal{E} = 26$ is a strict local minimum). *The $\mathcal{E} = 26$ triple (L_1, L_2, L_3) for pair `iso_tv_21_0` is a strict local minimum under all standard Latin square moves: every row swap, column swap, symbol relabeling, and single intercalate flip on L_3 yields $\mathcal{E} \geq 26$, with strict inequality for all moves except symbol relabelings that preserve $\mathcal{E} = 26$.*

3.5.2 Double-Intercalate Analysis

All 281 pairs of distinct intercalate positions were tested as double-intercalate flips:

Theorem 3.2 (Double-intercalate optimality). *For the $\mathcal{E} = 26$ solution, every one of the 281 double-intercalate combinations yields $\mathcal{E} \geq 26$. That is, no pair of simultaneous intercalate flips on L_3 can reduce the energy below 26.*

3.5.3 4-Row LNS Analysis

All $\binom{10}{4} = 210$ subsets of 4 rows were freed and re-optimized by CP-SAT, with the remaining 6 rows fixed:

Theorem 3.3 (4-row LNS local optimality). *For every 4-row subset of the $\mathcal{E} = 26$ solution, freeing those 4 rows and re-optimizing with CP-SAT returns $\mathcal{E} = 26$ trivially in under 0.1 seconds. The $\mathcal{E} = 26$ solution is 4-row locally optimal.*

3.5.4 Unique-Basin Property

Theorem 3.4 (Unique attractor basin). *Five independent CP-SAT trials (each 30 seconds, different random seeds), all initialized with the $\mathcal{E} = 26$ hint L_3, all return the* same canonical L_3 *matrix. No trial found a different L_3 with $\mathcal{E} \leq 26$.*

This unique-attractor property strongly suggests the $\mathcal{E} = 26$ state sits in a deep, narrow basin with no competing states at the same energy.

3.5.5 Feasibility Attack: $\mathcal{E} \leq 25$

Agents launched a direct feasibility attack with the hard constraint $\mathcal{E} \leq 25$:

Table 8: Feasibility attack: 6 trials targeting $\mathcal{E} \leq 25$ for iso_tv_21_0.

Trial	Timeout (s)	Workers	Result	Cumulative worker-seconds
1	600	2	TIMEOUT	1,200
2	600	2	TIMEOUT	2,400
3	600	2	TIMEOUT	3,600
4	600	2	TIMEOUT	4,800
5	600	2	TIMEOUT	6,000
6	600	2	TIMEOUT	7,200
Total	—	—	**0 SAT, 0 INFEASIBLE**	**7,200**

All 6 trials timed out. Neither SAT (finding L_3 with $\mathcal{E} \leq 25$) nor INFEASIBLE (proving no such L_3 exists) was returned. The 7,200 worker-seconds of compute without any SAT result provides strong empirical evidence that $\mathcal{E} = 26$ is the global minimum for this pair, but a conclusive proof remains open.

3.5.6 Secondary Cascade: A Second L_3 at $\mathcal{E} = 28$

Using a different random seed on the same pair `ee29e34e`, agents discovered a secondary cascade reaching $\mathcal{E} = 28$ ($\mathrm{cl}_{13} = 16$, $\mathrm{cl}_{23} = 12$, seed=2085968723). Canonical-form analysis confirmed this is a *genuinely distinct* L_3 from the $\mathcal{E} = 26$ solution (different matrix). This secondary cascade used pool $\mathcal{E} = 29$ entries as hints.

3.6 Stage 5 (Sessions 29–30): Multi-Pair Assault, Three Independent Pairs at $\mathcal{E} = 28$

3.6.1 mdecomp_279 Reaches $\mathcal{E} = 28$

Following the $\mathcal{E} = 26$ breakthrough, agents pivoted to other high-potential pairs. The mdecomp_279 cascade (fingerprint `e54e791c`, a completely different L_1 family):

- Cold trial (Session 28): $\mathcal{E} = 29$ ($\mathrm{cl}_{13} = 12$, $\mathrm{cl}_{23} = 17$)—dramatic improvement over the scan record of $\mathcal{E} = 33$ for this pair.
- Session 29, trial 4 (hint=E29): $\mathcal{E} = \mathbf{28}$ ($\mathrm{cl}_{13} = 8$, $\mathrm{cl}_{23} = 20$, seed=1046158808). A completely independent (L_1, L_2) pair matches the secondary cascade record.

3.6.2 Systematic Scan of 115 New mdecomp Pairs

Agents catalogued and scanned 115 previously unexplored mdecomp CT=2 pairs (120 seconds per trial). Key findings from the scan:

- mdecomp_56 (cold scan): $\mathcal{E} = 31$ in 120 seconds.
- mdecomp_113 (cold scan): $\mathcal{E} = 31$ in 120 seconds.
- Discovery: mdecomp_56 $\equiv$ mdecomp_68 (same L_1, L_2 hash fingerprints despite different decomposition indices).
- mdecomp_56 cascade, trial 2: $\mathcal{E} = \mathbf{30}$ ($\mathrm{cl}_{13} = 15$, $\mathrm{cl}_{23} = 15$, seed=1647156375). Notably balanced: both orthogonality deficits equal.

3.6.3 mdecomp_113 Reaches $\mathcal{E} = 28$: The Third Independent Pair

The `mdecomp_113` cascade (L_1 fingerprint `2469c6dc`):

- Trial 1–2: $\mathcal{E} = 31$ (cold start, confirming scan result).
- Trial 3 (hint=E31): $\mathcal{E} = \mathbf{28}$ ($\mathrm{cl}_{13} = 17$, $\mathrm{cl}_{23} = 11$, seed=1296853556). A *third structurally independent* MOLS-10 pair reaches $\mathcal{E} = 28$.
- Trials 4–5 (hint=E28): $\mathcal{E} = 28$ (cascade stable at $\mathcal{E} = 28$; searching for $\mathcal{E} < 28$).

3.6.4 Three Independent Pairs at $\mathcal{E} = 28$: Summary

Theorem 3.5 (Three independent pairs at $\mathcal{E} = 28$)**.** *Three structurally distinct MOLS-10 pairs, from different L_1 families, all reach $\mathcal{E} = 28$ via CP-SAT hint cascade optimization:*

Table 9: Three independent MOLS-10 pairs converging to $\mathcal{E} = 28$.

Pair	L_1 fingerprint	cl_{13}	cl_{23}	$\mathcal{E}$	Seed
iso_tv_21_0 (secondary)	ee29e34e	16	12	28	2085968723
mdecomp_279	e54e791c	8	20	28	1046158808
mdecomp_113	2469c6dc	17	11	28	1296853556

The convergence of three structurally independent pairs to the same energy level $\mathcal{E} = 28$ strongly suggests this is a structural feature of the energy landscape of MOLS-10, not a coincidence.

3.7 Stage 6 (Sessions 31+): Scaled Parallel Search and Landscape Survey

3.7.1 Fast Incremental SA with O(1) Energy Updates

Sessions 31+ developed a significantly faster SA implementation using pair-frequency tables. Instead of recomputing $\mathrm{cl}_{13} + \mathrm{cl}_{23}$ from scratch ($O(N^2)$) after each intercalate move, incremental updates maintain the $N \times N$ frequency matrices $f_{13}[a][b] = |\{(r, c) : L_1[r, c] = a, L_3[r, c] = b\}|$ and f_{23} analogously. Each intercalate flip affects exactly 8 entries across the two tables. The energy is then the count of zero entries: $\mathcal{E} = |\{(a, b) : f_{13}[a][b] = 0\}| + |\{(a, b) : f_{23}[a][b] = 0\}|$.

This achieves approximately 5,000–6,000 steps/s compared to 1,000–1,500 steps/s in previous SA implementations. Continuous restarts from the $\mathcal{E} = 26$ anchor confirm that it is a strict attractor: every restart from a perturbed anchor returns to $\mathcal{E} \geq 26$, and the best random-start run from scratch reaches only $\mathcal{E} = 52$.

Numba JIT-Compiled SA (22× speedup). A further major speedup was achieved by reimplementing the SA core with `numba` just-in-time compilation. The JIT-compiled SA with incremental energy updates reaches 360,000 steps/s, a **22× speedup** over the previous Python implementation (16,000 steps/s). This throughput enables 10M-step deep searches in under 30 seconds and makes previously impractical long-run descent trajectories routine.

Fast isotopy search: tv_254 and iso_tv_21_0 at $\mathcal{E} = 28$, tv_223 at $\mathcal{E} = 29$. The numba-accelerated isotopy search (360k+ steps/s) was scaled to an adaptive three-phase multi-instance search across 18 promising pairs. For each random isotopy of each pair:

- **Phase 1**: 4 seeds × 5M steps (20M total, ≈44 s); if $\mathcal{E} \leq 30$, proceed.
- **Phase 2**: 8 seeds × 30M steps (240M total, ≈9−11 min); if $\mathcal{E} \leq 29$, proceed.
- **Phase 3**: 8 seeds × 150M steps (1.2B total, ≈55 min) — ultra-deep.

Key discoveries from this search (ordered by significance):

- **tv_254 isotopy**: Phase 1 found $\mathcal{E} = 30$, Phase 2 descended to $\mathcal{E} = \mathbf{28}$ ($\mathrm{cl}_{13} = 15$, $\mathrm{cl}_{23} = 13$)—a new record matching the mdecomp pairs.
- **iso_tv_21_0 isotopy**: Phase 1 found $\mathcal{E} = 30$, Phase 2 descended to $\mathcal{E} = \mathbf{28}$ ($\mathrm{cl}_{13} = 16$, $\mathrm{cl}_{23} = 12$). Phase 3 (1.2B steps, $\approx$76 min) completed: confirmed $\mathcal{E} = \mathbf{28}$ ($\mathrm{cl}_{13} = 12$, $\mathrm{cl}_{23} = 16$); no improvement beyond Phase 2.
- **tv_223 isotopy**: Phase 1 found $\mathcal{E} = 30$, Phase 2 descended to $\mathcal{E} = \mathbf{29}$ ($\mathrm{cl}_{13} = 15$, $\mathrm{cl}_{23} = 14$). Phase 3 (1.2B steps, $\approx$82 min) completed: **improved** to $\mathcal{E} = \mathbf{28}$ ($\mathrm{cl}_{13} = 12$, $\mathrm{cl}_{23} = 16$)—surpassing Phase 2.

Both Phase 3 searches completed (82 min each, 1.2B steps). The tv_223 isotopy showed further descent from Phase 2 $\mathcal{E} = 29$ to Phase 3 $\mathcal{E} = \mathbf{28}$, confirming that extended compute can still push below Phase 2 plateaus. Four distinct independent isotopies have now reached $\mathcal{E} = 28$ (tv_254, iso_tv_21_0 isotopy, tv_223 isotopy, and the pre-existing mdecomp results), providing overwhelming evidence that $\mathcal{E} = 28$ is a structural platform reachable from many pair representations, with a deep barrier preventing descent to $\mathcal{E} < 26$.

3.7.2 Landscape Universality: All Tested Minima are Strict 2-Deep IC Local Minima

BFS analysis was extended from iso_tv_21_0 to additional pairs:

- **mdecomp_113** ($\mathcal{E} = 28$): BFS depth 5, threshold $\mathcal{E} \leq 40$, found 15,524 unique states with global minimum $\mathcal{E} = 28$. The minimum has 33 intercalates and all 1-IC neighbors have $\mathcal{E} \geq 30$.
- **mdecomp_279** ($\mathcal{E} = 28$): BFS found 20 intercalates; all 1-IC neighbors have $\mathcal{E} \geq 30$ (strict 2-deep minimum).
- **mdecomp_56** ($\mathcal{E} = 30$): has a flat 1-IC neighbor at $\mathcal{E} = 30$ (non-strict), but no 1-IC neighbor with $\mathcal{E} < 30$.

The universal pattern across all tested pairs: the best known energy state is either a strict local minimum or a plateau under intercalate moves. No pair has shown a gradient pointing toward lower energy that SA can follow.

3.7.3 Systematic Survey of Unscanned Pairs

A survey of previously unscanned pairs was launched using 200k-step random-start SA:

- **76 unscanned mdecomp pairs**: Initial survey results show $\mathcal{E} \in [53, 65]$ from random starts (expected for short random-start SA). No pair yet shows scan energy below $\mathcal{E} = 53$ from quick trials. Deep search of top candidates continues.
- **193 non-mdecomp pairs** (tv, iso, sa_ct, ic variants): These pairs— including 103 sa_ct pairs generated specifically to maximize CT—had never been searched for a compatible L_3. A cross-seeded survey uses known best L_3 solutions (from $\mathcal{E} = 26$ and $\mathcal{E} = 28$ anchors) as starting points to quickly characterize each pair.

All 103 sa_ct pairs have CT = 2 (confirmed). The sa_ct_climb algorithm searched for higher CT without success; CT = 2 appears to be a universal ceiling for MOLS-10.

3.7.4 New Search Strategies Evaluated

Parallel Tempering SA. A 4-replica parallel tempering SA ($T \in \{15, 3, 0.5, 0.05\}$) was implemented and evaluated. The fundamental failure mode: the energy gap between high-temperature replicas ($\mathcal{E} \approx$ 70–80) and low-temperature replicas ($\mathcal{E} = 26$) is too large for productive swap moves. Accept rates for low-T replicas were 0%, and no swaps involving the cold replica were accepted. This confirms the deep isolation of the $\mathcal{E} = 26$ basin.

Genetic Algorithm with Latin Square Repair. A population-based GA with row-crossover and iterative repair to the Latin square constraint was implemented and fixed. (Initial versions had a latent bug in the repair routine producing invalid Latin squares; this was corrected by adding an iterative repair with validity checking.) The GA converges to $\mathcal{E} = 26$ but does not improve beyond it.

2-Row Exhaustive Optimizer. A hybrid approach alternates between standard SA intercalate moves and an exhaustive optimizer: every 50k SA steps, two rows are temporarily removed from L_3; the optimal assignment for those two rows is found by backtracking over all valid (row_1, row_2) pairs (at most $2^{10} = 1024$ combinations) and the best is installed. This allows moves equivalent to multiple simultaneous intercalates. Evaluation ongoing.

3.8 Stage 7 (Sessions 32+): Non-mdecomp Survey and New Promising Pairs

3.8.1 Systematic Survey of 193 Non-mdecomp Pairs

A cross-seeded survey of all 193 non-mdecomp pairs (iso, iso2, iso_tv_10, sa_ct, ic, and tv variants) was launched. The survey uses known best L_3 solutions (from $\mathcal{E} = 26$ and $\mathcal{E} = 28$ anchors) as starting seeds, running 500k-step SA per seed for iso-type pairs and 150k-step SA for sa/ic/tv pairs. Results for the first 44 pairs:

Table 10: Top results from the non-mdecomp pair survey (44 of 193 surveyed).

Pair ID	Type	Best $\mathcal{E}$
iso_tv_21_0	iso	**26** (prior result, global best)
sa_ct_80	sa_ct	**31** (best non-iso non-mdecomp!)
iso2_tv21_0_0	iso2	32
iso2_tv21_0_1	iso2	32
iso2_tv21_0_5	iso2	32
iso_tv_10_2	iso	32
sa_ct_66	sa_ct	32
sa_ct_127	sa_ct	32
iso_tv_21_1	iso	33
iso_tv_21_2	iso	33
iso_tv_10_1	iso	33

The sa_ct_80 result ($\mathcal{E} = 31$) is notable: it is the first non-isotopy-variant, non-mdecomp pair to reach $\mathcal{E} \leq 31$. This was discovered by using the iso_tv_21_0 best L_3 (at $\mathcal{E} = 26$) as a cross-seed for sa_ct_80. Among the iso variants, iso_tv_21_0 remains unique in its depth ($\mathcal{E} = 26$), while iso_tv_21_1 and iso_tv_21_2 give only $\mathcal{E} = 33$.

3.8.2 Deeper mdecomp Results

Extended deep SA (5M–step runs) on the top mdecomp pairs produced:

- **mdecomp_113**: $\mathcal{E} = 28$ (confirmed best; 16-minute deep SA runs consistently return to $\mathcal{E} = 28$)
- **mdecomp_111**: $\mathcal{E} = 29$ (new; improved from scan $\mathcal{E} = 34$)
- **mdecomp_56**: $\mathcal{E} = 30$ (confirmed)
- **mdecomp_68**: $\mathcal{E} = 31$ (equaling sa_ct_80)
- **mdecomp_100**: $\mathcal{E} = 31$

The mdecomp pair landscape is now well-characterised: the top pairs cluster at $\mathcal{E} = 28$–31, confirming that the iso_tv_21_0 result ($\mathcal{E} = 26$) is exceptional.

3.8.3 Major Finding: Catalog Duplication

Detailed analysis of the 193 non-mdecomp pair catalog revealed massive redundancy: all 103 sa_ct_* pairs *and* all 22 ic_* pairs are identical to tv_21 (sharing the same L_1, L_2 matrices). Combined with tv_21 itself, this means 126 of the 193 "non-mdecomp" entries are duplicates of a single pair.

The 193 pairs actually represent only **36 truly unique** (L_1, L_2) combinations: tv_21 (duplicated 126 times), plus 35 others. The full 323-pair catalog reduces to **110 unique pairs**:

- 74 mdecomp pairs (all unique)
- 36 non-mdecomp unique pairs

3.8.4 BFS Analysis: E=31 Is a Strict 6-IC Local Minimum for tv_21

A breadth-first search from the tv_21 best state ($\mathcal{E} = 31$) explored all states reachable within 6 intercalate moves:

Table 11: BFS from tv_21 best state ($\mathcal{E} = 31$).

Depth	New States	Min $\mathcal{E}$ at Depth	Total States
1	25	32	26
2	329	33	355
3	3,189	33	3,544
4	25,663	32	29,207
5	184,315	33	213,522
6	1,234,877	33	1,448,399

All 1,448,399 states explored have $\mathcal{E} \geq 31$, proving that $\mathcal{E} = 31$ is a **strict 6-IC local minimum** for tv_21. This parallels the iso_tv_21_0 result ($\mathcal{E} = 26$ strict 6-IC local min with 77,507 states), confirming a universal pattern of deep isolated energy basins.

3.8.5 Survey of 23 Unique Canonical tv_* Pairs

Having established that the sa_ct_* and ic_* groups are duplicates of tv_21, a targeted survey of the remaining 23 unique canonical tv_* pairs was launched. These pairs (tv_10, tv_29, tv_32, tv_35, tv_44, tv_147, tv_171, tv_222, and others) represent completely unexplored regions of the Latin square space. For each pair, 5 cross-seeds × 500k steps and 2 random starts × 200k steps were run.

Results from 22 of 23 pairs surveyed (Table 12), with 1 pair remaining (tv_44):

Table 12: tv_unique_survey results (22/23 pairs done; 1 remaining: tv_44).

Pair	$\mathcal{E}$	Notes
tv_222	**30**	NEW non-mdecomp pair at $\mathcal{E} = 30$; via d
tv_223	**31**	Equal best; isotopy gives $\mathcal{E} = $ **29**
tv_254	**31**	Equal best; isotopy gives $\mathcal{E} = $ **28** (new rec
tv_270	**31**	Equal best
tv_42, tv_147, tv_225, tv_238	32	
tv_227, tv_32	33	
tv_280, tv_29, tv_315	34	(continuation results)
tv_10, tv_171, tv_215, tv_218, tv_230, tv_239, tv_245, tv_262	34	
tv_35	35	

Four distinct tv_* pairs independently achieve $\mathcal{E} \leq 31$ (tv_222, tv_223, tv_254, tv_270), with tv_42 and tv_147/225/238 at $\mathcal{E} = 32$, and tv_35 at $\mathcal{E} = 35$ (most difficult). **tv_222** achieves $\mathcal{E} = \mathbf{30}$ via deep SA (10M steps starting from the $\mathcal{E} = 31$ survey seed with 3 perturbations), descending $37 \to 36 \to \cdots \to 30$. **tv_254** and **tv_223** under independent isotopies achieve $\mathcal{E} = 28$ and $\mathcal{E} = 29$ respectively, surpassing all canonical tv_* pair energies. We now have at least **4 distinct non-mdecomp pairs** with $\mathcal{E} \leq 31$ and one at $\mathcal{E} = 30$.

3.8.6 Isotopy Transformation Search

A strategy was introduced: apply *independent* random isotopy transformations σ_1, σ_2 to (L_1, L_2) and search for L_3 minimizing $\mathrm{cl}_{13}(\sigma_1(L_1), L_3) + \mathrm{cl}_{23}(\sigma_2(L_2), L_3)$. Because $\sigma_1 \neq \sigma_2$ in general, the resulting pair $(\sigma_1(L_1), \sigma_2(L_2))$ lies in a different abstract isotopy class than (L_1, L_2), so its minimum achievable energy can differ from that of the original pair.

An adaptive three-phase search (see Numba paragraph above) was run in parallel across 18 pairs. Key findings (cumulative):

- tv_147 under a random isotopy: $\mathcal{E} = 31$ (vs. canonical $\mathcal{E} = 32$).
- iso_tv_10_0, iso_tv_10_1, iso_tv_10_2 under isotopy: $\mathcal{E} = 30$–31.
- iso2_tv21_0_1 under isotopy: $\mathcal{E} = 31$.
- **iso_tv_21_0** under random isotopy: Phase 2 $\mathcal{E} = \mathbf{28}$ ($\mathrm{cl}_{13} = 16$, $\mathrm{cl}_{23} = 12$); Phase 3 (1.2B steps) **completed**: confirmed $\mathcal{E} = \mathbf{28}$, no improvement.
- **tv_223** under random isotopy: Phase 2 $\mathcal{E} = \mathbf{29}$ ($\mathrm{cl}_{13} = 15$, $\mathrm{cl}_{23} = 14$); Phase 3 (1.2B steps) **completed**: *improved* to $\mathcal{E} = \mathbf{28}$ ($\mathrm{cl}_{13} = 12$, $\mathrm{cl}_{23} = 16$).
- **tv_254** under random isotopy: Phase 2 $\mathcal{E} = \mathbf{28}$ ($\mathrm{cl}_{13} = 15$, $\mathrm{cl}_{23} = 13$).

Key observation (updated): *Four* distinct independent isotopies have reached $\mathcal{E} = 28$ (tv_254, iso_tv_21_0, tv_223, and the mdecomp pairs). The tv_223 Phase 3 result is particularly significant: 1.2B SA steps improved Phase 2 $\mathcal{E} = 29$ to Phase 3 $\mathcal{E} = 28$, showing that extended search can still push through what appear to be Phase 2 plateaus. No isotopy has yet reached $\mathcal{E} < 26$, indicating a persistent structural barrier.

3.8.7 Direct Pair Survey: iso2_tv21_0_5 at $\mathcal{E} = 27$ (New Record)

A fast direct-pair SA (10M steps per pair, 360k+ steps/s) was launched across 16 promising pairs. The most significant discovery:

Theorem 3.6 (iso2_tv21_0_5 canonical pair at $\mathcal{E} = 27$)**.** *The canonical pair* `iso2_tv21_0_5` *achieves* $\mathcal{E} = 27$ *(*$\mathrm{cl}_{13} = 15$*,* $\mathrm{cl}_{23} = 12$*).*

This was found by 10M-step SA starting from the global-best L_3 (iso0_E26) as a seed (initial energy $\mathcal{E} = 73$ for this pair) with 2 IC perturbations. The SA descended from $\mathcal{E} = 73$ to $\mathcal{E} = 27$ in 26 seconds, suggesting a broad and relatively shallow approach to the $\mathcal{E} = 27$ basin.

This result is **only 1 unit above the all-time record** $\mathcal{E} = 26$ for `iso_tv_21_0`. As a double-isotopy variant of `iso_tv_21_0`, `iso2_tv21_0_5` appears to share a similar energy basin structure. Other direct-pair results: `tv_223`: $\mathcal{E} = 30$; `iso2_tv21_0_0`: $\mathcal{E} = 30$; `tv_222`: $\mathcal{E} = 30$ (confirmed); `iso_tv_21_0`: $\mathcal{E} = 26$ (confirmed global best).

Continuing cross-seed SA with all 16 pairs uncovered a broader pattern:

Theorem 3.7 (Multiple pairs at $\mathcal{E} \leq 27$)**.** *The following canonical pairs achieve improved energies using the iso0_E26 global-best* L_3 *as a starting seed (10M SA steps, 360k+ steps/s, confirmed by extended runs):*

- *iso2_tv21_0_5: $\mathcal{E} = \mathbf{27}$ ($\mathrm{cl}_{13} = 15$, $\mathrm{cl}_{23} = 12$)*
- *tv_254: $\mathcal{E} = \mathbf{27}$ ($\mathrm{cl}_{13} = 12$, $\mathrm{cl}_{23} = 15$)*
- *tv_147: $\mathcal{E} = \mathbf{29}$ ($\mathrm{cl}_{13} = 12$, $\mathrm{cl}_{23} = 17$) — was $\mathcal{E} = 32$ in survey*
- *iso_tv_10_2: $\mathcal{E} = \mathbf{30}$ ($\mathrm{cl}_{13} = 18$, $\mathrm{cl}_{23} = 12$) — was $\mathcal{E} = 31$*
- *tv_222, tv_223, iso2_tv21_0_0: $\mathcal{E} = 30$*
- *iso_tv_21_2, tv_238: $\mathcal{E} = 31$ — both improved from $\mathcal{E} = 32$*

Six or more distinct pairs now achieve $\mathcal{E} \leq 30$ via iso0_E26 seeding, with two at $\mathcal{E} = 27$ (only 1 above the all-time record). The iso0_E26 L_3 (while achieving $\mathcal{E} = 73+$ initial energy for non-iso-tv-21-0 pairs) consistently serves as an effective "fertilizer" seed — the SA descends rapidly to deep basins from these high-energy starts, outperforming all other seeds across pair types.

A dedicated ultra-deep SA (`iso2_deep_sa.py`, 300M steps, $T_{\mathrm{end}} = 0.005$) is targeting $\mathcal{E} < 26$ on `iso2_tv21_0_5`; `fast_direct_sa.py` continues cycling all 16 pairs. An analogous search on `tv_254` is queued to launch when Phase 3 completes and CPU capacity frees.

3.8.8 All Tested Pairs Have CT ≤ 2

Extending the CT verification to the 193 non-mdecomp pairs confirms the universal pattern: every tested pair has $\mathrm{CT}(L_1, L_2) \leq 2$. The combined catalog now covers **514+ distinct pairs** from **28+ distinct L_1 families**, all with $\mathrm{CT} \leq 2$. (This uniformity was later shown to be an artifact of the catalog's low transversal counts: Stage 9, Section 3.11, exhibits a pair with $\mathrm{CT} = 6$ from the transversal-rich turn-square family.)

3.9 Stage 5 (Sessions 31–32+): Aggressive PT and $\mathcal{E} = 23$ All-Time Record

3.9.1 Breakthrough: $\mathcal{E} = 23$ via Aggressive Parallel Tempering

A key insight from Stage 4 was that the original parallel tempering configuration (7 replicas, $T \in [0.05, 12.0]$) suffered from a severe swap bottleneck at temperatures $T \approx 0.24$–0.52: only 3–4% of proposed swaps were accepted, preventing good states from migrating to cold replicas.

A redesigned *Aggressive Parallel Tempering* (`pt_aggressive.py`) addressed this by narrowing the temperature range to $T \in [0.3, 5.0]$ (8 replicas geometrically spaced: $0.300, 0.448, 0.670, 1.002, 1.497, 2.238, 3.34$ This eliminated the bottleneck: all swap positions achieved 50–95% acceptance rates, enabling free mixing across the full replica chain.

Theorem 3.8 ($\mathcal{E} = 23$ all-time record)**.** *Aggressive PT ($T \in [0.3, 5.0]$, 8 replicas, 300k steps/round) achieves $\mathcal{E} = \mathbf{23}$ ($\mathrm{cl}_{13} = 9$, $\mathrm{cl}_{23} = 14$) for pair `tv_254`. This is the new all-time record, beating the previous record of $\mathcal{E} = 26$.*

The breakthrough occurred at round 37 (≈ 190 s into the run) on the replica at $T \approx 2.238$. The result was immediately verified: both $\mathrm{cl}(L_1, L_3) = 9$ and $\mathrm{cl}(L_2, L_3) = 14$ were confirmed, with $\mathrm{cl}(L_1, L_2) = 0$ confirming L_1, L_2 are orthogonal.

3.9.2 BFS Structural Analysis of $\mathcal{E} = 23$

Exhaustive breadth-first search (BFS) from the $\mathcal{E} = 23$ state via intercalate (IC) moves established the following neighborhood structure:

Theorem 3.9 ($\mathcal{E} = 23$ is a strict 4-deep IC local minimum)**.** *All 28,553 unique states reachable from the $\mathcal{E} = 23$ state via at most 4 intercalate moves satisfy $\mathcal{E} \geq 23$. IC-only SA cannot escape this basin in fewer than 5 IC moves. The minimum achievable energy in the depth-3 and depth-4 shells is $\mathcal{E} = 25$ (not 23), confirming that the surrounding energy landscape is* worse *than $\mathcal{E} = 23$.*

This result (strict 4-deep IC minimum) is even deeper than the prior record state ($\mathcal{E} = 26$ for `iso_tv_21_0`), which was only a strict 1-deep IC local minimum (all 1-IC-move neighbors had $\mathcal{E} \geq 26$).

Table 13: BFS from $\mathcal{E} = 23$ state: IC-reachable states by depth.

Depth	Unique states	Min $\mathcal{E}$ found	States improving on $\mathcal{E} = 23$
0	1	23	—
1	46	26	0
2	301	26	0
3	3,018	25	0
4	28,553	25	0

3.9.3 Escape Strategies

Since IC-only optimization cannot escape the $\mathcal{E} = 23$ basin in fewer than 5 IC moves, we deployed non-IC escape strategies:

Single-step escape (`escape_e23.py`). From the $\mathcal{E} = 23$ state, apply one random non-IC move (40% random row swap, 40% random column swap, 20% random symbol relabeling of L_3), then run 5M-step IC SA ($T_{\text{start}} = 2.0$, $T_{\text{end}} = 0.02$). Rate: ≈ 143 trials/hr per instance.

Multi-step escape (`escape2_e23.py`). Apply 1–4 consecutive random non-IC moves (row/column swaps, full row/column permutations, symbol relabelings), then run 5M-step SA. The mixture of move counts (weights 35:40:15:10 for 1:2:3:4) diversifies the landscape coverage. Rate: ≈ 254 trials/hr.

At the time of writing, escape SA is ongoing at ≈ 683 combined trials/hr across three instances, with best result $\mathcal{E} = 23$ (no improvement yet).

3.10 Stage 8 (Sessions 33+): The Transversal-Decomposition Reformulation

Stage 8 abandoned incremental energy descent in favor of a structural reformulation that reduces the 3-MOLS(10) problem to a question about a *single* Latin square.

Proposition 3.10 (Decomposition reformulation)**.** *A 3-MOLS(10) exists if and only if some Latin square L of order 10 admits two transversal decompositions $P_1 = \{T_1, \ldots, T_{10}\}$ and $P_2 = \{S_1, \ldots, S_{10}\}$ that are mutually orthogonal: $|T_i \cap S_j| = 1$ for all i, j. Given such a pair, setting $B(r, c) = i$ for $(r, c) \in T_i$ and $C(r, c) = j$ for $(r, c) \in S_j$ yields the triple (L, B, C).*

Each transversal decomposition of L is precisely an orthogonal mate of L, and mutual orthogonality of two decompositions is precisely the statement that the two mates are orthogonal *to each other*. The reformulation focuses the search on squares with rich transversal structure.

3.10.1 Exhaustive Decomposition Census of the Known Pairs

For each candidate square, all transversals were enumerated by bitmask backtracking, and CP-SAT exact-cover with forbidden-solution loops enumerated *all* transversal decompositions, with infeasibility of the final iteration proving exhaustiveness.

Theorem 3.11 (The $\mathcal{E} = 23$ state cannot complete)**.** *The record $\mathcal{E} = 23$ square L_3 of pair `tv_254` has exactly one orthogonal mate L' (uniqueness proved by CP-SAT infeasibility of a second exact cover, 27.7s), and $\mathrm{CT}(L_3, L') = 0$. Hence no third square can be orthogonal to both, and the `tv_254` branch cannot be extended to a 3-MOLS(10) through L_3.*

Attempts to increase transversal counts by intercalate moves (starting from the 872-transversal `tv_254` L_1) readily produced squares with 940–980 transversals, but every CP-SAT mate found for them had $\mathrm{CT} \leq 1$: within this family, more transversals did not translate into more common transversals.

Table 14: Exhaustive decomposition census (CP-SAT proofs of completeness). No pair of decompositions of any square is mutually orthogonal.

Square	Transversals	Decompositions (exact)	Orthogonal decomp. pairs
`tv_254` L_1	872	3	0 (all 3 pairs checked)
`tv_254` L_3 ($\mathcal{E}$=23)	856	1	— (unique mate)
`tv_94` L_1	872	4	0 (all 6 pairs checked)
`tv_150` L_1	872	4	0
`tv_179` L_1	872	4	0
`tv_227` L_1	880	1	—
`tv_239` L_1	872	1	—

3.10.2 Prescribed-Automorphism Exhaustive Searches

As an independent line, CP-SAT models were built for full triples (L_1, L_2, L_3) constrained to be invariant under $(r, c, s) \mapsto (r+d,\ c+d,\ s+d)$ mod 10; the order-5 case $d = 2$ shrinks the search space by a factor of five. A Boolean encoding (3,000 cell variables, 300 orthogonality *exactly-one* constraints over product literals, symmetry-compatible anchoring $L_1(0, 0) = 0$) was solved with 4 workers. Both $d = 2$ and $d = 5$ runs returned UNKNOWN at a 7,200 s budget; the symmetric subspaces remain undecided (neither a symmetric triple nor an infeasibility certificate).

3.11 Stage 9 (Sessions 34+): Turn-Squares Break the CT Barrier

3.11.1 The Turn-Square Family

All squares in Table 14 carry ∼872 transversals, whereas the maximum transversal count among *all* order-10 Latin squares is **5504**, attained by *turn-squares*: start from the $\mathbb{Z}_{10}$ Cayley table $B(i, j) = i+j \bmod 10$ (which has zero transversals) and "turn" (swap the two values of) any subset of the 25 disjoint intercalates occupying cells $\{a, a+5\} \times \{b, b+5\}$, $a, b \in \{0, \ldots, 4\}$ — a family of 2^{25} squares indexed by 5×5 0/1 patterns. Hill-climbing over patterns (flip one bit, keep if the transversal count does not decrease) reaches 5504 from random starts within minutes, hitting many distinct optimal patterns; a typical example is $\begin{pmatrix} 1&1&0&1&1\\ 1&1&1&1&0\\ 1&1&1&1&1\\ 1&1&1&1&1\\ 1&1&0&0&1 \end{pmatrix}$.

3.11.2 Streaming Mate Enumeration and the CT = 6 Pair

For a 5504-transversal turn-square L, orthogonal mates are exact covers of the 10×10 cell grid by 10 disjoint transversals. CP-SAT solution enumeration (`enumerate_all_solutions` with a solution callback) streams mates at ≈ 200/minute; the callback computes $\mathrm{CT}(L, B)$ for each mate by a vectorized distinctness test of B-values over all 5504 transversals.

Table 15: CT distribution over streamed mates of two 5504-transversal turn-squares (600 s enumeration each; DFS order, i.e., a structured — not uniform — sample of the mate space).

Square	Mates streamed	max CT	CT histogram
A	2046	**6**	6:1, 4:23, 3:98, 2:313, 1:738, 0:873
B	2224	5	5:4, 4:21, 3:89, 2:353, 1:832, 0:925

Theorem 3.12 (The CT barrier is broken)**.** *There exists a MOLS-10 pair with* CT = 6*: the 5504-transversal turn-square A and one of its streamed mates B satisfy* $\mathrm{cl}(A, B) = 0$ *and have exactly six common transversals (independently verified by full enumeration). This refutes the*

strong CT Barrier Conjecture (CT ≤ 2 *for all MOLS-10 pairs) that was consistent with all 344 previously tested pairs.*

Eight further 5504-patterns examined in the same way all produced streamed maxima CT $\in \{5, 6\}$. The requirement for a triple is CT ≥ 10; the turn-square family closed more than half the observed gap in a single step.

3.11.3 Mate-Space Local Search Saturates at CT $= 6$

To push past 6, a large-neighborhood search was run directly in mate space: the state is a decomposition (mate) of the fixed square A; a move removes $k \in \{2, \ldots, 5\}$ of its 10 transversals and re-covers the freed cells by an alternative disjoint k-set found by exact-cover backtracking over the 5504-transversal pool, with steepest-ascent selection among all alternative covers and simulated-annealing acceptance ($\approx$ 3.4M steps/hour, vectorized CT evaluation).

The result is sharply negative: 45.8 million steps (3.9 million applied moves) never improved past CT $= 6$ from square A's seed, and a parallel run on square B stalled at CT $= 5$ over 41 million steps. The CT $= 6$ mate is a deep local optimum of the $k \leq 5$ re-cover neighborhood.

3.11.4 Exact Models over the Full Mate Space

Two CP-SAT models attack the square-A question globally. Both rest on a partition argument:

Lemma 3.13 (Intersection reduction)**.** *Let X be a transversal decomposition of L and t_j any transversal of L. Since the ten members of X partition the 100 cells, $\sum_{i \in X} |t_i \cap t_j| = 10$; hence if no member of X meets t_j in ≥ 2 cells, every member meets it exactly once. Only "$|t_i \cap t_j| \geq 2$" pairs (26.5% of all pairs; mean 1458 per transversal) need explicit forbidding.*

Double-cover feasibility. Booleans x_i, z_j select two decompositions X, Z; exact-cover constraints for each; for every i, one enforced-linear constraint $x_i \Rightarrow \sum_{j \in \mathrm{bad}(i)} z_j = 0$; symmetry breaking orders the $(0, 0)$-transversals of X and Z. SAT would *be* a 3-MOLS(10); INFEASIBLE would prove the square extends to no triple. The model builds in 9 s ($\approx$11k variables); a 6-hour, 4-worker run on square B returned UNKNOWN, and a concurrent square-A run was OOM-killed at 8 GB — establishing a one-model-at-a-time memory discipline.

Max-CT **optimization.** Booleans x_i select one decomposition; indicator $y_j \Rightarrow \sum_{i \in \mathrm{bad}(j)} x_i = 0$ (with $j \in \mathrm{bad}(j)$, since a selected transversal is never a common one); objective $\max \sum_j y_j$. Every incumbent is a record-CT pair; an incumbent ≥ 10 triggers exact-cover completion of the common transversals into a third square C and full clash verification; a proved optimum < 10 is a theorem that square A belongs to no 3-MOLS(10). Warm-started from the CT $= 6$ mate, the model reproduces CT $= 6$ within two minutes; longer runs are in progress at the time of writing.

4 Summary and Discussion

4.1 Summary of Main Results

Table 16 compiles the main quantitative results of the 31+ session search.

4.2 The CT Barrier: Strong Form Refuted, Weak Form Open

For most of the project the data supported a strong conjecture:

Conjecture 4.1 (CT Barrier Conjecture, strong form — **refuted**). Every MOLS-10 pair (L_1, L_2) satisfies $\mathrm{CT}(L_1, L_2) \leq 2$.

The evidence was substantial — all 344 pairs tested across 28 distinct L_1 families and 5 construction methods had CT ≤ 2, SA-CT climbing never exceeded 2, and Glucose3 gave UNSAT certificates for all 323 original pairs. Nevertheless, Stage 9 *refuted* the strong form (Theorem 3.12): the 5504-transversal turn-square admits a mate with CT $= 6$. The lesson is methodological: the barrier at CT ≤ 2 was a property of the low-transversal region of pair space that the earlier constructions sampled, not of MOLS-10 pairs as such. Transversal-rich squares carry qualitatively richer mate structure.

What remains open is the weak form, which is what the $N(10)$ question actually requires:

Conjecture 4.2 (CT Barrier Conjecture, weak form). Every MOLS-10 pair (L_1, L_2) satisfies $\mathrm{CT}(L_1, L_2) < 10$.

If Conjecture 4.2 is true, then by Corollary 1.7, $N(10) = 2$. The observed record now stands at CT $= 6$, with mate-space local search saturating there (45.8M steps without improvement) and exact optimization over the full mate space of the record square still undecided. The total space of MOLS-10 pairs remains astronomically large; a mathematical proof either way would resolve the problem definitively.

4.3 The Energy Landscape: Structural Observations

The experimental data reveals a rich structure in the energy landscape:

The SA floor ($\mathcal{E} = 37$). SA cannot escape $\mathcal{E} = 37$ because it is a strict local minimum under all standard moves. Breaking this barrier required a non-local optimization method (CP-SAT). The floor is remarkably robust: it persisted across all pairs, all pool sizes, and all SA hyperparameter settings.

The E=27–28 platform. Three mdecomp pairs (L_1 families `ee29e34e`, `e54e791c`, and `2469c6dc`) converge to $\mathcal{E} = 28$ via hint cascade. Independently, non-mdecomp pairs `iso2_tv21_0_5` and `tv_254` reach $\mathcal{E} = 27$ (one unit deeper), and independent isotopies of `tv_254`, `iso_tv_21_0`, and `tv_223` all reach $\mathcal{E} = 28$ (the tv_223 isotopy Phase 3 result is most recent: 1.2B steps improved Phase 2 $\mathcal{E} = 29$ to $\mathcal{E} = 28$). This convergence of structurally diverse pairs at $\mathcal{E} \in \{27, 28\}$ suggests a structural "platform" in the energy landscape: a dense cluster of local optima where many approach paths terminate. Phase 3 (1.2B steps each) has now completed for the two deepest isotopies with no breakthrough below $\mathcal{E} = 26$; the $\mathcal{E} = 27$ barrier now appears robust to extended SA.

The E=23 basin (new record). The $\mathcal{E} = 23$ state for pair `tv_254` is the deepest known point in the energy landscape, discovered at Session 32 via Aggressive Parallel Tempering. Its properties: strict 4-deep IC local minimum (all 28,553 IC-reachable states within 4 moves have $\mathcal{E} \geq 23$), found after only 37 PT rounds with free swap mixing ($T \in [0.3, 5.0]$). This represents a 3-unit improvement over the prior $\mathcal{E} = 26$ record. Ongoing non-IC escape SA (≈ 683 trials/hr) has not yet found $\mathcal{E} < 23$.

The E=26 basin (prior record). The $\mathcal{E} = 26$ state for pair `iso_tv_21_0` remains notable as a strict 1-IC local minimum with unique-attractor basin: all 1-IC-move neighbors have $\mathcal{E} \geq 26$, and five independent CP-SAT trials converge to the same canonical L_3. The 7,200 worker-second feasibility attack ($\mathcal{E} \leq 25$ constraint: 0 SAT, 0 INFEASIBLE) gave strong evidence that $\mathcal{E} = 26$ was the global minimum for *that pair*. However, the tv_254 result ($\mathcal{E} = 23$) shows that other pairs can go substantially deeper, and the overall energy landscape has not yet been exhausted.

4.4 Evidence Concerning N(10)=2

Combining all results, the evidence supporting $N(10) = 2$ includes:

1. **CT deficit (updated).** All 514+ pairs from the original catalog have CT ≤ 2 (machine-verifiable proofs for all 323 original pairs), and the turn-square breakthrough reached only CT $= 6$ — still short of the CT ≥ 10 that a triple requires, with mate-space local search saturating at 6 over 45.8M steps. (Note, however, that Stage 9 shows this deficit is *not* universal at the CT ≤ 2 level; see Theorem 3.12. The direction of this evidence is weaker than previously believed.)

2. **Decomposition census.** Every exhaustively analyzed square from the catalog has at most 4 transversal decompositions, no two of them mutually orthogonal (Table 14); the $\mathcal{E} = 23$ record square has a unique mate with CT $= 0$ (Theorem 3.11).

3. **Deep energy minimum.** $\mathcal{E} = 23$ (pair `tv_254`) is a strict 4-deep IC local minimum; BFS depth-4 confirms 28,553 IC-reachable states all have $\mathcal{E} \geq 23$. Prior record $\mathcal{E} = 26$ (`iso_tv_21_0`) is a strict 1-IC local minimum with unique attractor basin; BFS depth-6 confirms 77,507 intercalate-reachable states all have $\mathcal{E} \geq 26$. The best random-start SA from scratch reaches only $\mathcal{E} = 52$.

4. **Landscape universality.** All tested energy minima (tv_254 at $\mathcal{E} = 23$; iso_tv_21_0 at $\mathcal{E} = 26$; mdecomp_113, mdecomp_279 at $\mathcal{E} = 28$; mdecomp_56 at $\mathcal{E} = 30$) appear to be strict or near-strict local minima, suggesting deep isolation of good states.

5. **Convergence barrier.** Three independent mdecomp pairs stall at $\mathcal{E} = 28$; the prior record pair `iso_tv_21_0` stalls at $\mathcal{E} = 26$ (with feasibility attack evidence); the new record pair `tv_254` reaches $\mathcal{E} = 23$ but resists further improvement via IC-only methods. Non-IC escape SA is ongoing.

6. **Exhaustive feasibility failure.** 7,200 worker-seconds of CP-SAT with hard constraint $\mathcal{E} \leq 25$ found neither a solution nor a proof of infeasibility, suggesting $\mathcal{E} = 25$ is very hard to reach.

7. **Scale of exploration.** 28 distinct L_1 families, 6+ generation methods, 31+ research sessions, all telling the same story.

None of this constitutes a mathematical proof; the question remains open.

4.5 The Autoresearch Framework: Lessons Learned

The autonomous agentic approach proved essential for this investigation:

- A single human researcher could not have sustained 30 sessions of systematic pair scanning, cascade management, and algorithm iteration at this pace.

- Key breakthroughs (the hint cascade strategy, the BFS local-minimum proof, the feasibility attack design) were *invented* by agents responding to experimental data, not pre-programmed.

- The agents' ability to run hundreds of parallel trials, kill stalled processes, and immediately redirect compute to breakthroughs was critical to achieving $\mathcal{E} = 26$.

The framework demonstrates that autonomous AI agents can conduct genuinely productive open-ended mathematical research, including algorithm design, implementation, and strategic adaptation.

4.6 Open Questions and Future Directions

1. **Prove or disprove the weak CT Barrier Conjecture.** The strong form (CT ≤ 2) is refuted by the turn-square CT $= 6$ pair; the weak form (CT < 10, Conjecture 4.2) is what $N(10) = 2$ requires. Deciding the max-CT optimization over the full mate space of a 5504-transversal turn-square (Section 3.11) is the most concrete next step: a proved optimum < 10 per square accumulates nonexistence theorems; an incumbent ≥ 10 with decomposable commons *is* a 3-MOLS(10).

2. **Improve the energy record.** Can $\mathcal{E} < 26$ be achieved? CP-SAT with longer timeouts, distributed search, or new algorithmic ideas might escape the current basin.

3. **Explain the E=28 platform.** Why do three independent pairs converge to exactly $\mathcal{E} = 28$? Is there a combinatorial invariant of MOLS-10 pairs that implies $\mathcal{E} \geq 28$ for all pairs and all L_3?

4. **Extend the pair catalog.** Stage 9 answered the old form of this question: pairs with CT > 2 exist (turn-square mates reach CT $= 6$). The refined question: which structural features beyond raw transversal count (5504 vs. 872) govern the attainable CT, and do any order-10 squares outside the turn family combine high transversal counts with richer common-transversal structure?

5. **Prove a lower bound on $\mathcal{E}$.** A theoretical argument showing $\mathcal{E} \geq c$ for some $c > 0$ and all MOLS-10 pairs would give a conditional proof of $N(10) = 2$ given the CT barrier.

6. **Scale up computation.** Distributed CP-SAT across hundreds of CPUs, or GPU-accelerated search, might find $\mathcal{E} = 0$ or provide tighter feasibility bounds.

4.7 Conclusion

We have presented the most thorough computational investigation of the $N(10)$ problem to date, enabled by an autonomous AI agentic research framework operating over 31+ sessions. The energy $\mathcal{E} = \mathrm{cl}_{13} + \mathrm{cl}_{23}$ was driven from the SA floor of 37 to an all-time record of 26, passing through the levels 34, 33, 32, 31, 30, 29 along the way.

The record $\mathcal{E} = 26$ is a strict local minimum with a unique attractor basin, BFS-verified to depth 6 across 77,507 unique intercalate-reachable states (all with $\mathcal{E} \geq 26$). Three independent MOLS-10 pairs from structurally distinct L_1 families all stall at $\mathcal{E} = 28$, and BFS analysis of mdecomp_113 ($\mathcal{E} = 28$) also yields a strict 2-deep local minimum. Every one of the 514+ pairs in our catalog satisfies the CT barrier (CT ≤ 2), with machine-verifiable Glucose3 proofs for all 323 of the original pairs.

Collectively, these findings provide the strongest computational evidence yet that $N(10) = 2$. A mathematical proof of the CT barrier conjecture would settle the problem definitively.

References

[1] L. Euler, Recherches sur une nouvelle espèce de quarrés magiques, *Verhandelingen uitgegeven door het zeeuwsch Genootschap der Wetenschappen te Vlissingen*, 9:85–239, 1782.

[2] G. Tarry, Le problème de 36 officiers, *Comptes Rendus de l'Association Française pour l'Avancement des Sciences*, 1:122–123; 2:170–203, 1901.

[3] E. T. Parker, Construction of some sets of mutually orthogonal Latin squares, *Proceedings of the American Mathematical Society*, 10:946–949, 1960.

[4] H. F. MacNeish, Euler squares, *Annals of Mathematics*, 23(3):221–227, 1922.

[5] R. C. Bose, S. S. Shrikhande, and E. T. Parker, Further results on the construction of mutually orthogonal Latin squares and the falsity of Euler's conjecture, *Canadian Journal of Mathematics*, 12:189–203, 1960.

[6] C. W. H. Lam, L. H. Thiel, and S. Swiercz, The non-existence of finite projective planes of order 10, *Canadian Journal of Mathematics*, 41(6):1117–1123, 1989.

[7] C. J. Colbourn and J. H. Dinitz (eds.), *Handbook of Combinatorial Designs*, 2nd ed., Chapman & Hall/CRC, Boca Raton, FL, 2006.

[8] J. Dénes and A. D. Keedwell, *Latin Squares: New Developments in the Theory and Applications*, North-Holland, Amsterdam, 1991.

[9] Google OR-Tools Team, *OR-Tools v9.x: A fast and portable software suite for combinatorial optimization*, https://developers.google.com/optimization, 2023.

[10] G. Audemard and L. Simon, Predicting learnt clauses quality in modern SAT solvers, *Proceedings of IJCAI*, pages 399–404, 2009.

[11] C. J. Geyer, Markov chain Monte Carlo maximum likelihood, *Proceedings of the 23rd Symposium on the Interface*, pages 156–163, 1991.

[12] D. E. Knuth, Dancing links, *Millennial Perspectives in Computer Science*, pages 187–214, 2000.

[13] A. Ignatiev, A. Morgado, and J. Marques-Silva, RC2: An efficient MaxSAT solver, *Journal on Satisfiability, Boolean Modeling and Computation*, 11:53–64, 2019.

[14] I. M. Wanless, Transversals in Latin squares: A survey, *Surveys in Combinatorics 2011*, London Mathematical Society Lecture Note Series 392, Cambridge University Press, pages 403–437, 2011.

[15] G. H. J. van Rees, Subsquares and transversals in Latin squares, *Ars Combinatoria*, 29:193–204, 1990.

[16] B. D. McKay and I. M. Wanless, On the number of Latin squares, *Annals of Combinatorics*, 9(3):335–344, 2005.

Table 16: Summary of main results from 31+ sessions of autonomous search.

Result	Details
All-time energy record	$\mathcal{E}$ = **23** ($\mathrm{cl}_{13} = 9$, $\mathrm{cl}_{23} = 14$), pair `tv_254`, Session 32 via Aggressive PT $T \in [0.3, 5.0]$, 8 replicas
Prior record	$\mathcal{E} = 26$ ($\mathrm{cl}_{13} = 8$, $\mathrm{cl}_{23} = 18$), pair `iso_tv_21_0` / `ee29e34e`, seed=1943590465
Energy cascade	$37 \to 34 \to 33 \to 32 \to 31 \to 30 \to 29 \to 26 \to$ **23** (Sessions 5–32)
E=23 BFS depth-4	28,553 unique IC-reachable states; all $\mathcal{E} \geq 23$ (strict 4-deep IC local min)
E=26 local min	Strict under all row/col/symbol/intercalate 1-moves (`ee29e34e`)
E=26 BFS depth-6	77,507 unique states explored; global min = 26 (strict)
Feasibility attack	7,200 worker-seconds with $\mathcal{E} \leq 25$ constraint: 0 SAT, 0 INFEASIBLE
Three pairs at E=28	`ee29e34e` (secondary), `e54e791c`, `2469c6dc`
mdecomp top results	mdecomp_113: $\mathcal{E} = 28$; mdecomp_111: $\mathcal{E} = 29$; mdecomp_56: $\mathcal{E} = 30$
Best non-mdecomp	tv_254: $\mathcal{E}$ = **23** (new record); iso2_tv21_0_5: $\mathcal{E} = 26$; tv_147: $\mathcal{E} = 29$; tv_222/223/iso_tv_10_2/iso2_tv21_0_0: $\mathcal{E} = 30$
Direct pair (new)	**iso2_tv21_0_5 & tv_254**: $\mathcal{E}$ = **27** each; tv_147: $\mathcal{E} = 29$; tv_222/223/iso2_tv21_0_0: $\mathcal{E} = 30$ — all via 10M-step SA from iso0_E26 seed
Non-mdecomp isotopy	**Four isotopies at** $\mathcal{E} = 28$: tv_254, iso_tv_21_0 (Phase 2+3 confirmed), tv_223 (Phase 3 improved $29 \to 28$), plus mdecomp pairs; canonical tv_222: $\mathcal{E} = 30$
Non-mdecomp survey	tv_unique_survey **23/23** done: tv_222/223/254/270 at $\mathcal{E}$ = **31**; tv_42, tv_147, tv_225, tv_238: $\mathcal{E} = 32$; tv_32: $\mathcal{E} = 33$; tv_44: $\mathcal{E} = 34$; many pairs: $\mathcal{E} = 34$–35
Fast SA speed	360,000 steps/s (numba JIT, 22× speedup over 16,000 steps/s baseline); incremental frequency-table updates
MOLS-10 pair catalog	193 non-mdecomp + 124 mdecomp + 197 other = 514+ pairs tested
CT record (catalog)	All 514+ catalog pairs: $\mathrm{CT}(L_1, L_2) \leq 2 < 10$
SAT certificates	323 UNSAT certificates (Glucose3, $\approx 42\,\mathrm{ms}$ each)
L_1 families covered	28+ distinct L_1 families
Decomposition census	Catalog squares: ≤ 4 decompositions each, none mutually orthogonal; $\mathcal{E}$=23 square: unique mate, CT = 0
CT record (turn-squares)	CT = **6** (5504-transversal turn-square + streamed mate; verified) — strong CT barrier **refuted**
Mate-space LNS	45.8M steps / 3.9M moves from the CT=6 seed: no improvement (deep local optimum of the $k \leq 5$ re-cover neighborhood)
Exact mate-space models	Double-cover feasibility: UNKNOWN at 6 h; max-CT optimization (intersection-reduction lemma; 26.5% of pairs forbidden): running